

BUSINESS REPORT – MACHINE LEARNING-2 **PROJECT**

Casestudy - **EasyVisa**

Submitted by: Goli Bhagyasri

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PROBLEM STATEMENT:

Context:

Business communities in the United States are facing high demand for human resources, but one of the constant challenges is identifying and attracting the right talent, which is perhaps the most important element in remaining competitive. Companies in the United States look for hard-working, talented, and qualified individuals both locally as well as abroad.

The Immigration and Nationality Act (INA) of the US permits foreign workers to come to the United States to work on either a temporary or permanent basis. The act also protects US workers against adverse impacts on their wages or working conditions by ensuring US employers' compliance with statutory requirements when they hire foreign workers to fill workforce shortages. The immigration programs are administered by the Office of Foreign Labor Certification (OFLC).

OFLC processes job certification applications for employers seeking to bring foreign workers into the United States and grants certifications in those cases where employers can demonstrate that there are not sufficient US workers available to perform the work at wages that meet or exceed the wage paid for the occupation in the area of intended employment.

Objective

In FY 2016, the OFLC processed 775,979 employer applications for 1,699,957 positions for temporary and permanent labor certifications. This was a nine percent increase in the overall number of processed applications from the previous year. The process of reviewing every case is becoming a tedious task as the number of applicants is increasing every year.

The increasing number of applicants every year calls for a Machine Learning based solution that can help in shortlisting the candidates having a higher chance of VISA approval. OFLC has hired the firm EasyVisa for data-driven solutions. You, as a data scientist at EasyVisa, have to analyze the data provided and, with the help of a classification model:

Facilitate the process of visa approvals. Recommend a suitable profile for the applicants for whom the visa should be certified or denied based on the drivers that significantly influence the case status.

Data Description

The data contains the different attributes of the employee and the employer. The detailed data dictionary is given below.

- case_id: ID of each visa application
- continent: Information of continent the employee
- education_of_employee: Information of education of the employee
- has_job_experience: Does the employee have any job experience? Y= Yes; N = No
- requires_job_training: Does the employee require any job training? Y = Yes; N = No
- no_of_employees: Number of employees in the employer's company
- yr_of_estab: Year in which the employer's company was established

- `region_of_employment`: Information of foreign worker's intended region of employment in the US.
- `prevailing_wage`: Average wage paid to similarly employed workers in a specific occupation in the area of intended employment. The purpose of the prevailing wage is to ensure that the foreign worker is not underpaid compared to other workers offering the same or similar service in the same area of employment.
- `unit_of_wage`: Unit of prevailing wage. Values include Hourly, Weekly, Monthly, and Yearly.
- `full_time_position`: Is the position of work full-time? Y = Full-Time Position; N = Part-Time Position
- `case_status`: Flag indicating if the Visa was certified or denied

Exploratory Data Analysis:

Problem statement:

Facilitate the process of visa approvals. Recommend a suitable profile for the applicants for whom the visa should be certified or denied based on the drivers that significantly influence the case status.

Data Analysis:

- Below table shows first five rows of data

	case_id	continent	education_of_employee	has_job_experience	requires_job_training	no_of_employees	yr_of_estab	region_of_employment
0	EZYV01	Asia	High School	N	N	14513	2007	West
1	EZYV02	Asia	Master's	Y	N	2412	2002	Northeast
2	EZYV03	Asia	Bachelor's	N	Y	44444	2008	West
3	EZYV04	Asia	Bachelor's	N	N	98	1897	West
4	EZYV05	Africa	Master's	Y	N	1082	2005	South

FIG.1 : Head of data

- Below table shows last five rows of data

```
data.tail()
```

	case_id	continent	education_of_employee	has_job_experience	requires_job_training	no_of_employees	yr_of_estab	region_of_employment
25475	EZYV25476	Asia	Bachelor's	Y	Y	2601	2008	Sc
25476	EZYV25477	Asia	High School	Y	N	3274	2006	North
25477	EZYV25478	Asia	Master's	Y	N	1121	1910	Sc
25478	EZYV25479	Asia	Master's	Y	Y	1918	1887	V
25479	EZYV25480	Asia	Bachelor's	Y	N	3195	1960	Midw

FIG.2 : Tail of data

- **Data has 25480 rows and 12 columns**
- no_of_employees, yr_of_estab and prevailing_wage are the only numerical columns while the rest are categorical columns
- There are no null values in the data
- There are no duplicates or missing values in the data.

Statistical summary of data:

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
case_id	25480	25480	EZYV25480	1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
continent	25480	6	Asia	16861	NaN	NaN	NaN	NaN	NaN	NaN	NaN
education_of_employee	25480	4	Bachelor's	10234	NaN	NaN	NaN	NaN	NaN	NaN	NaN
has_job_experience	25480	2	Y	14802	NaN	NaN	NaN	NaN	NaN	NaN	NaN
requires_job_training	25480	2	N	22525	NaN	NaN	NaN	NaN	NaN	NaN	NaN
no_of_employees	25480.000	NaN	NaN	NaN	5667.043	22877.929	-26.000	1022.000	2109.000	3504.000	602069.000
yr_of_estab	25480.000	NaN	NaN	NaN	1979.410	42.367	1800.000	1976.000	1997.000	2005.000	2016.000
region_of_employment	25480	5	Northeast	7195	NaN	NaN	NaN	NaN	NaN	NaN	NaN
prevailing_wage	25480.000	NaN	NaN	NaN	74455.815	52815.942	2.137	34015.480	70308.210	107735.513	319210.270
unit_of_wage	25480	4	Year	22962	NaN	NaN	NaN	NaN	NaN	NaN	NaN
full_time_position	25480	2	Y	22773	NaN	NaN	NaN	NaN	NaN	NaN	NaN
case_status	25480	2	Certified	17018	NaN	NaN	NaN	NaN	NaN	NaN	NaN

Fig 3: Statistical summary of data

Observations

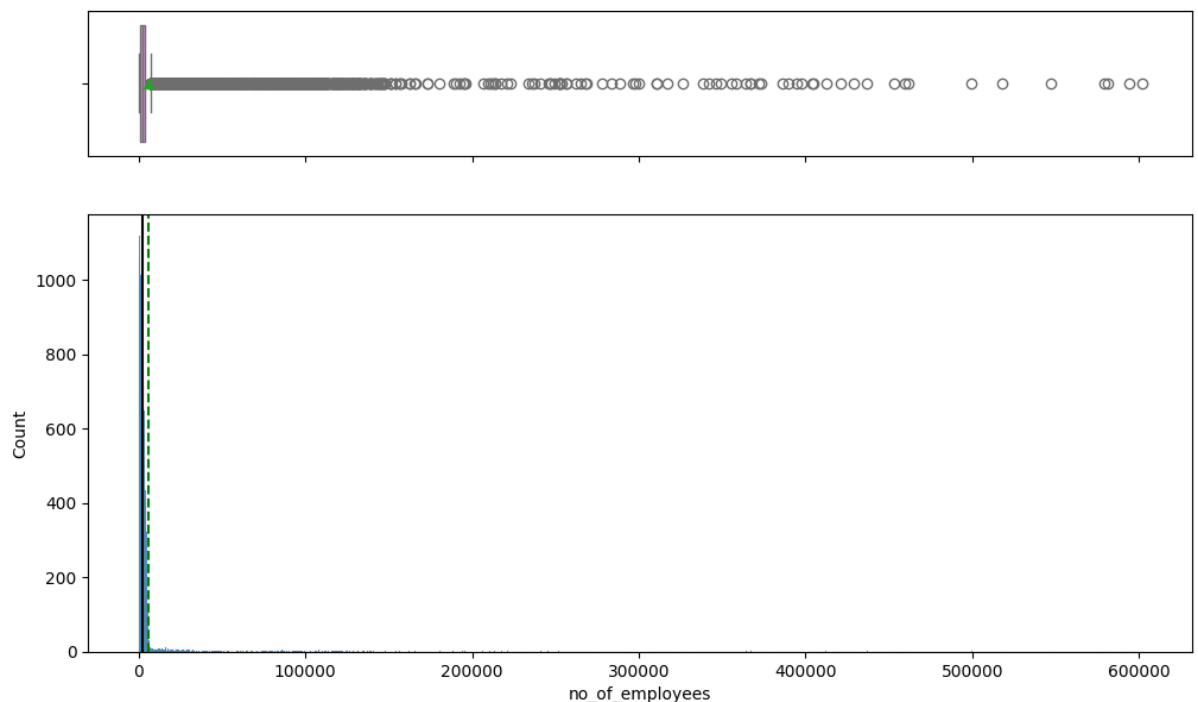
- Asia is the most frequent country from which employees are selected. However it seems to be skewed data.
- Employee with Bachelor's education are more in number
- Majority have a prior experience as 14802 people out of 25480 people.
- Most of the people donot need job training
- Northeast is the top region of employment
- Majority of them are working in full time position
- no_of_employees show negative values which is an error that need to be treated.

Observing unique values in the variables:

case_id:	25480 unique values
Continent:	6 unique values
education_of_employee:	4 unique values
has_job_experience:	2 unique values
requires_job_training:	2 unique values
region_of_employment:	5 unique values
unit_of_wage:	4 unique values
full_time_position:	2 unique values
case_status:	2 unique values

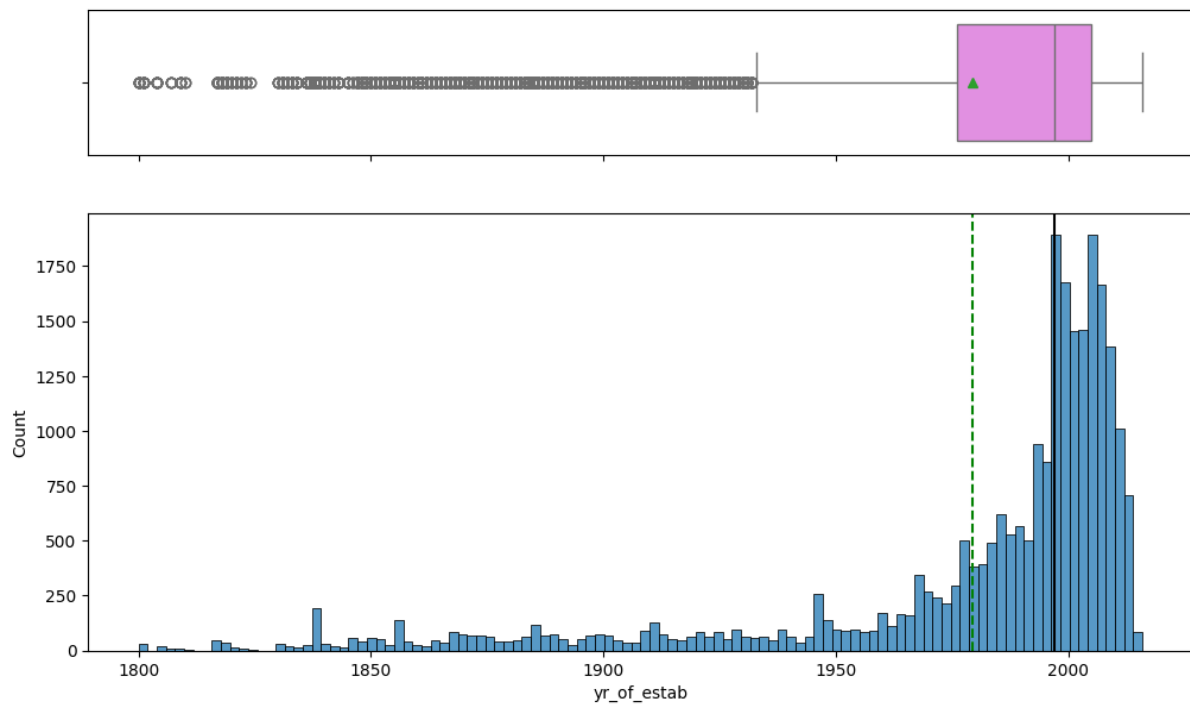
Univariate Analysis:

1. Observations on no_of_employees



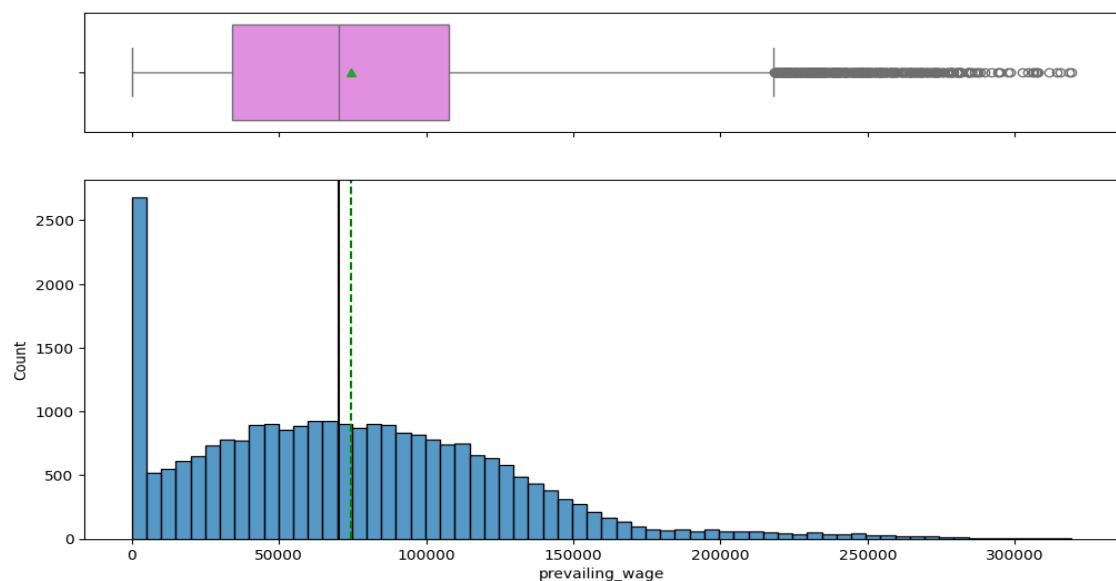
- The box is tightly aligned to the left, while the whiskers and outliers extend far to the right.
- This indicates extreme positive skewness—a small number of companies have massive employee counts.
- Several outlier are beyond the upper whisker, indicating negative values.

2. Observations on yr_of_estab



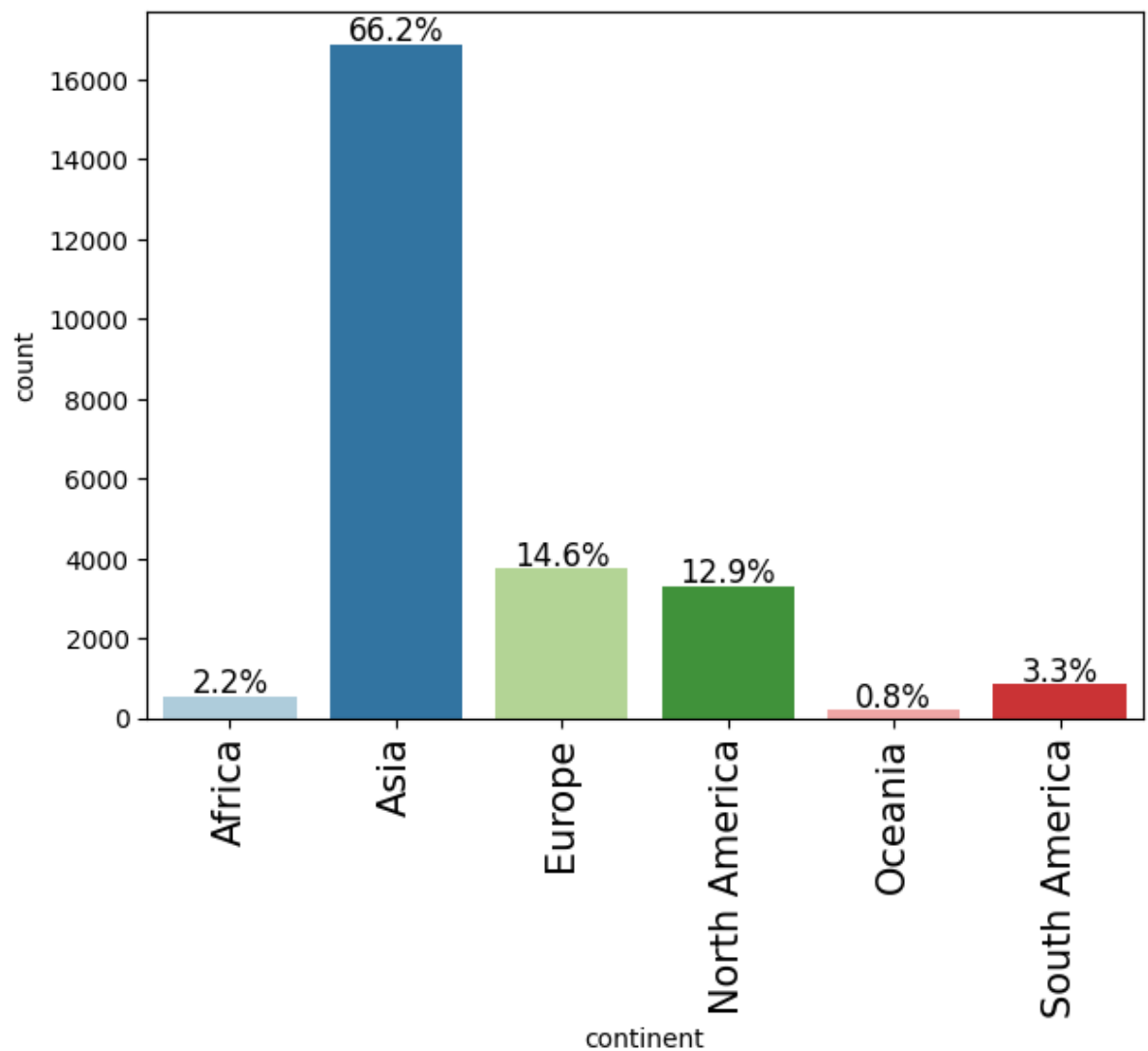
- Graph shows a heavily right skewed data, indicating that most of companies are established around 1980 to 2010 year

3. Observations on prevailing_wage

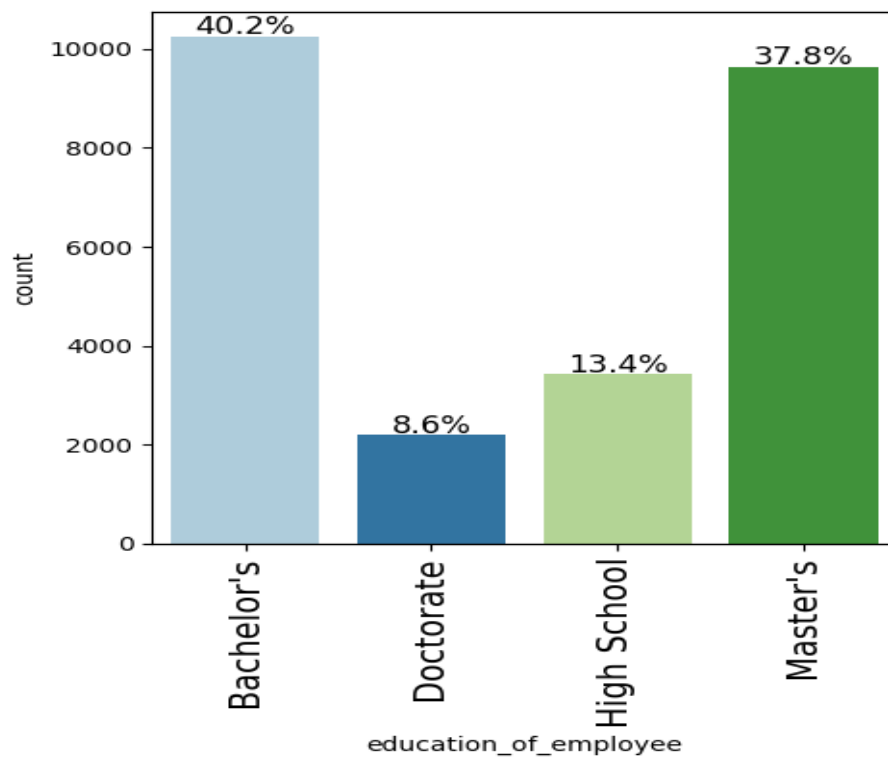


- The graph shows right skewed data, representing that most of the prevailing wages of other employees in same field are in moderate range.

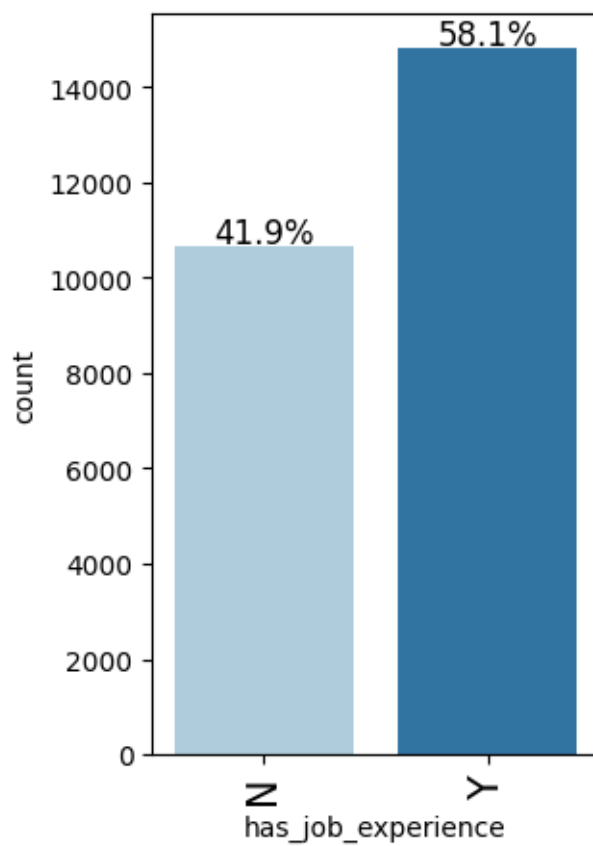
4. Observations on continent



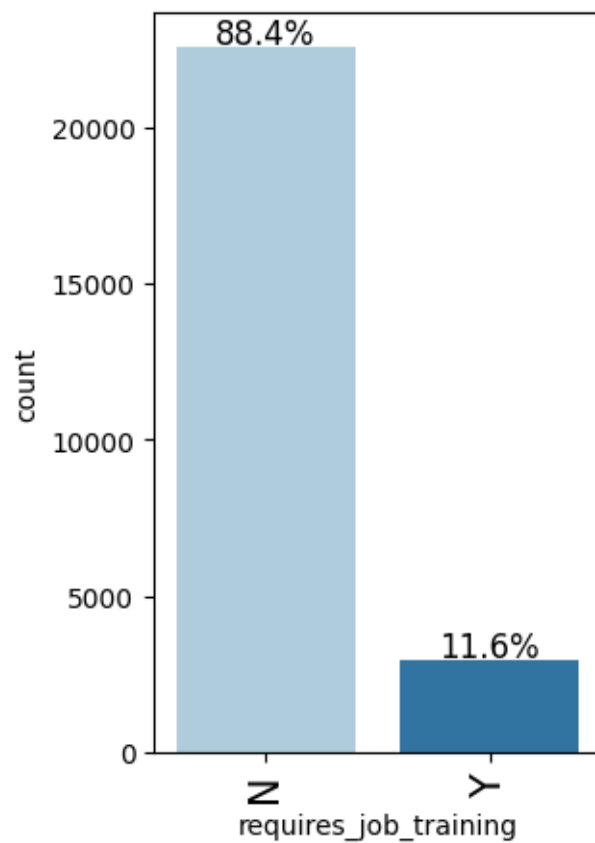
5. Observations on education_of_employee



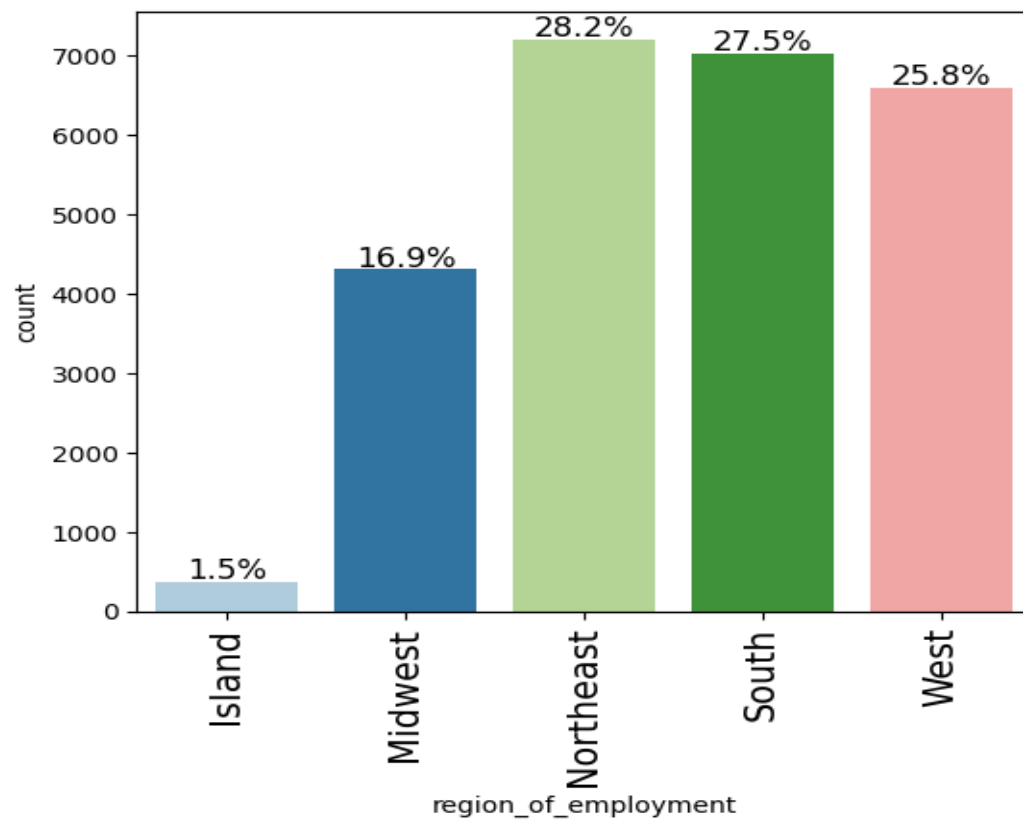
6. Observations on has_job_experience



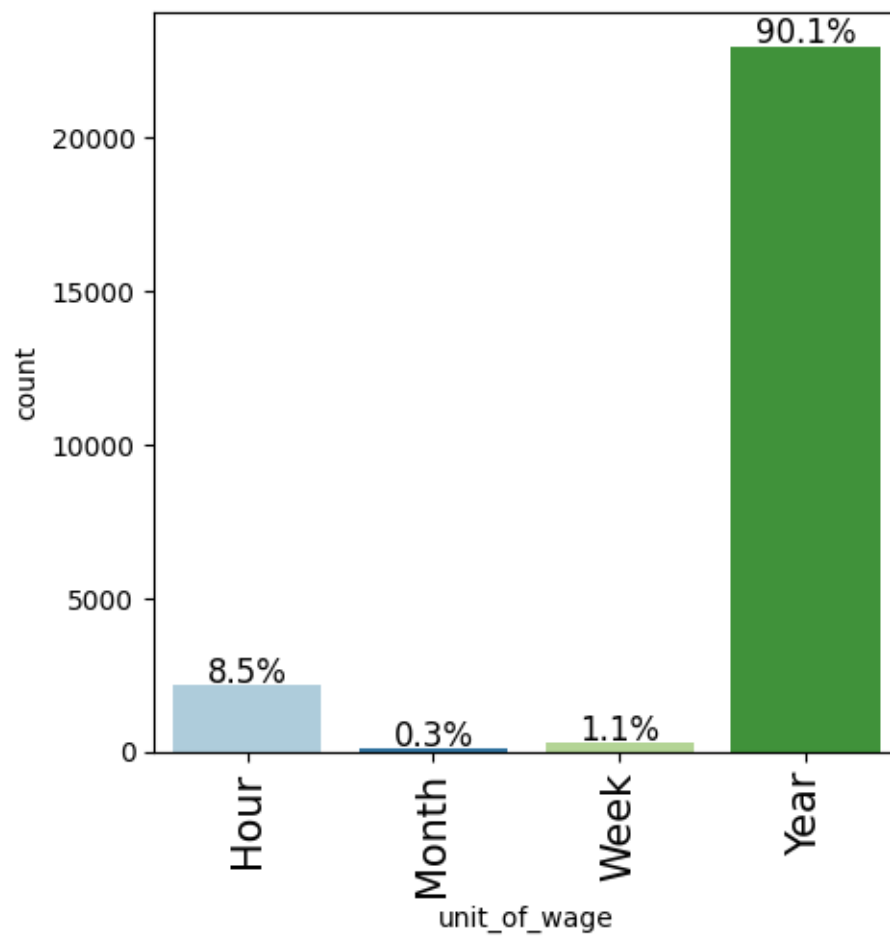
7. Observations on requires_job_training



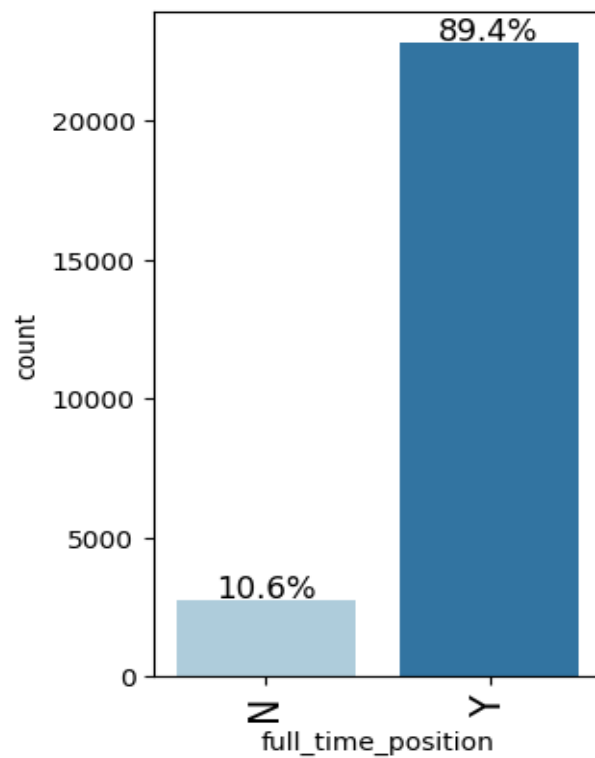
8. Observations on region_of_employment



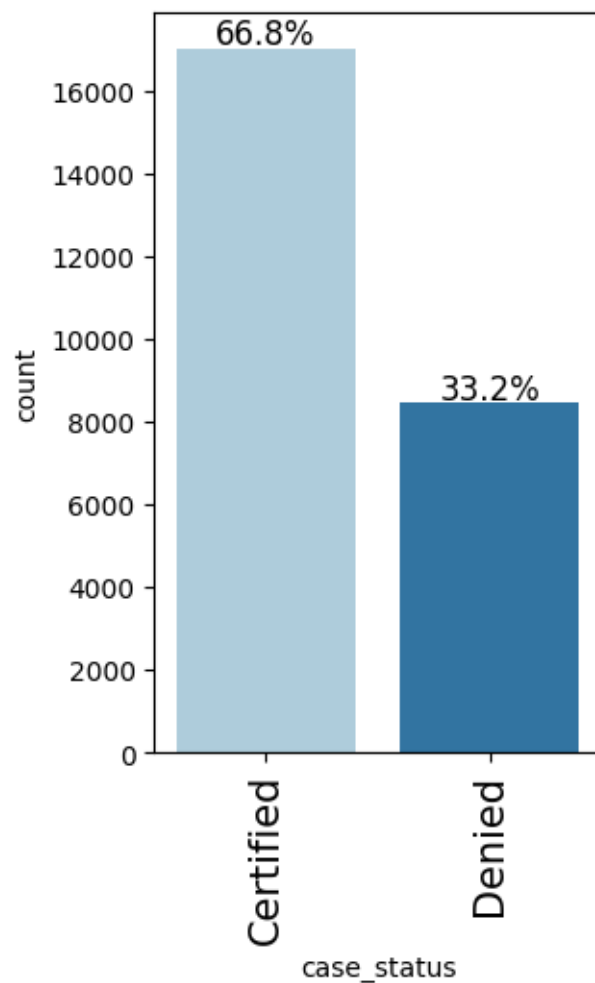
9. Observations on unit_of_wage



10. Observations on full_time_position



11. Observations on case_status

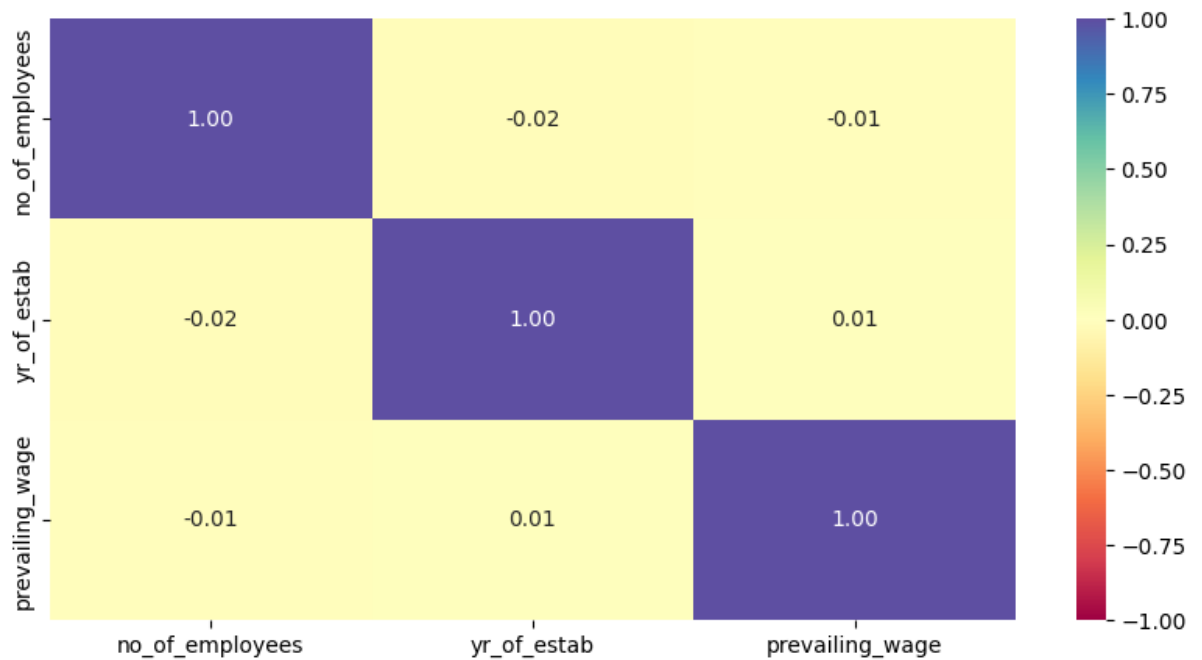


Observations:

- Asians are the most applied people when compared with other countries by 66.2%
- Majority of the applicants have Bachelor's and Master's degree i.e., around 40%.
- 58% of applicants have no prior job experience.
- 88.4% of applicants say they don't need a on job training
- Most of the applicants are interested to work and get placed in Northeast, South and West regions
- Employees get paid on yearly basis
- Majority of them i.e., around 90% are full time employed.
- Around 67% of the applicant visa's are certified or accepted and 33% are denied.

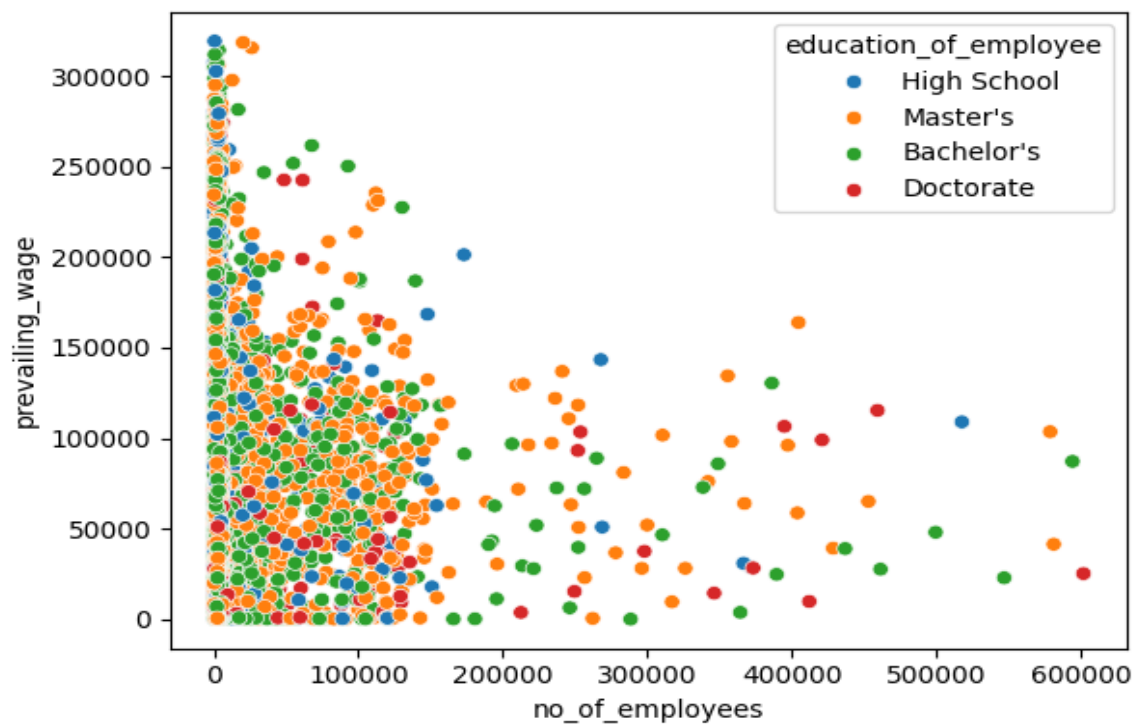
Bivariate Analysis:

1. Correlation Analysis

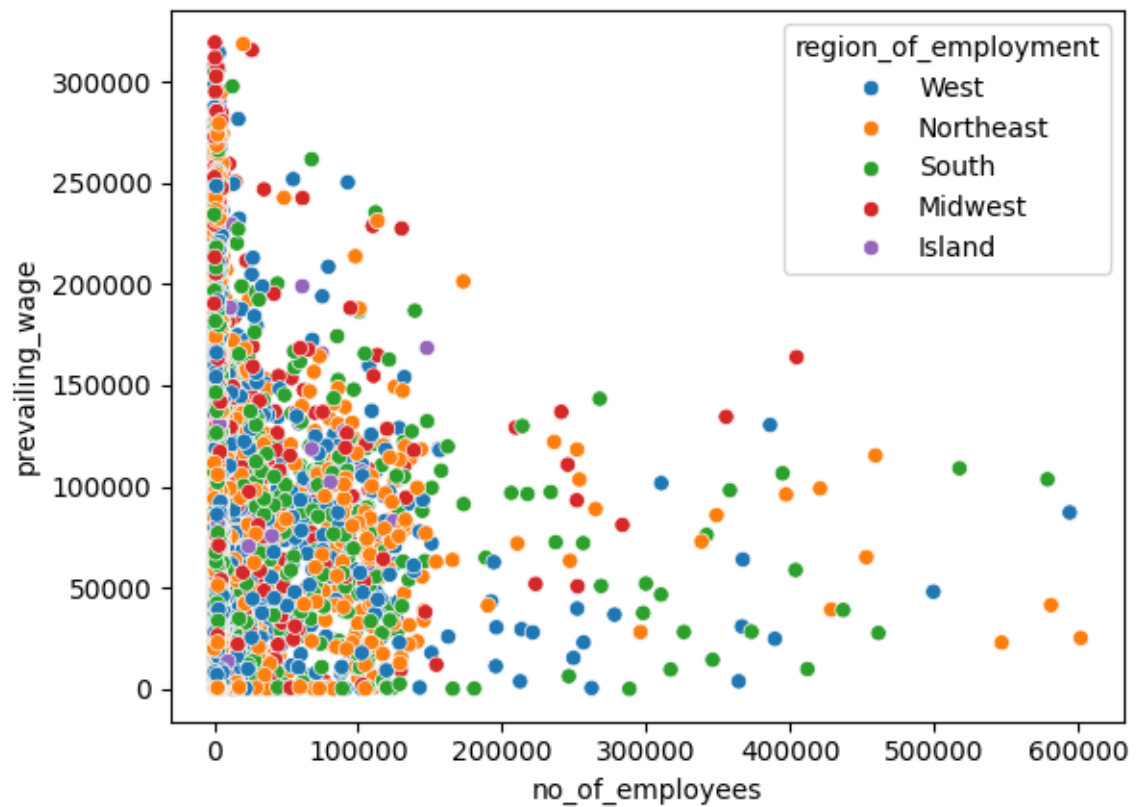


- No strong linear correlations exist between any of the three variables
- This states that changes in one variable do not predict or explain changes in another.

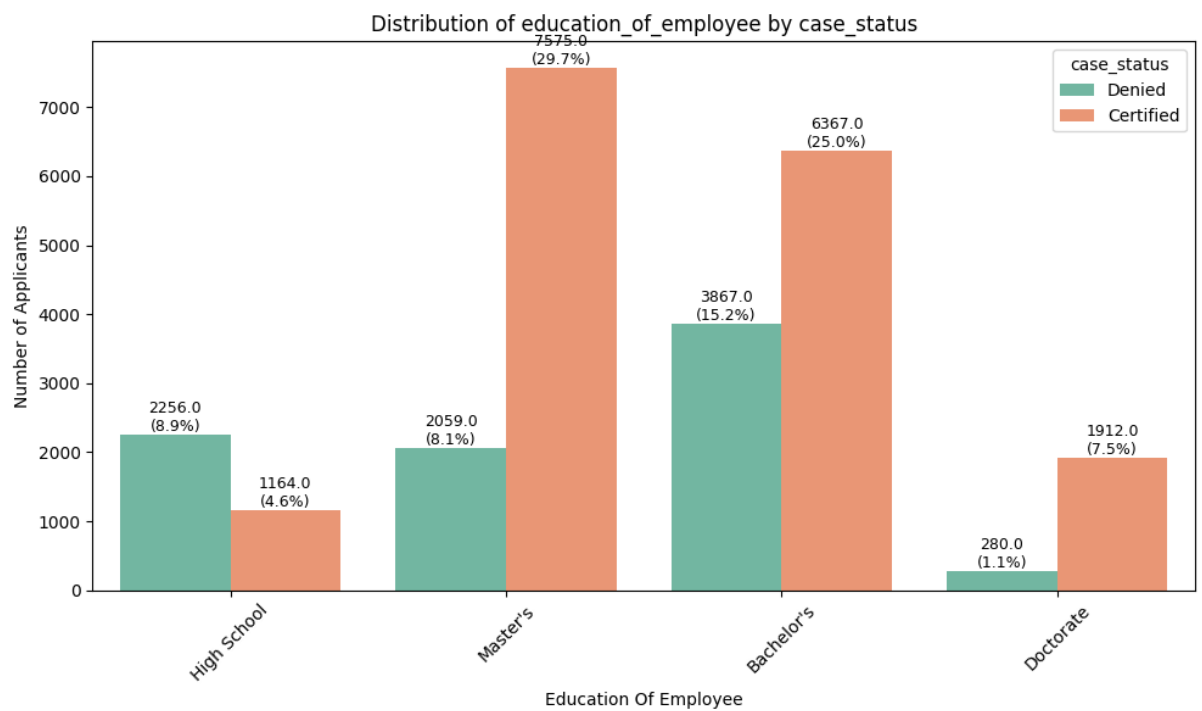
2. Observations on no_of_employees, prevailing_wage and education_of_employee



3. Observations on no_of_employees, prevailing_wage and region_of_employment

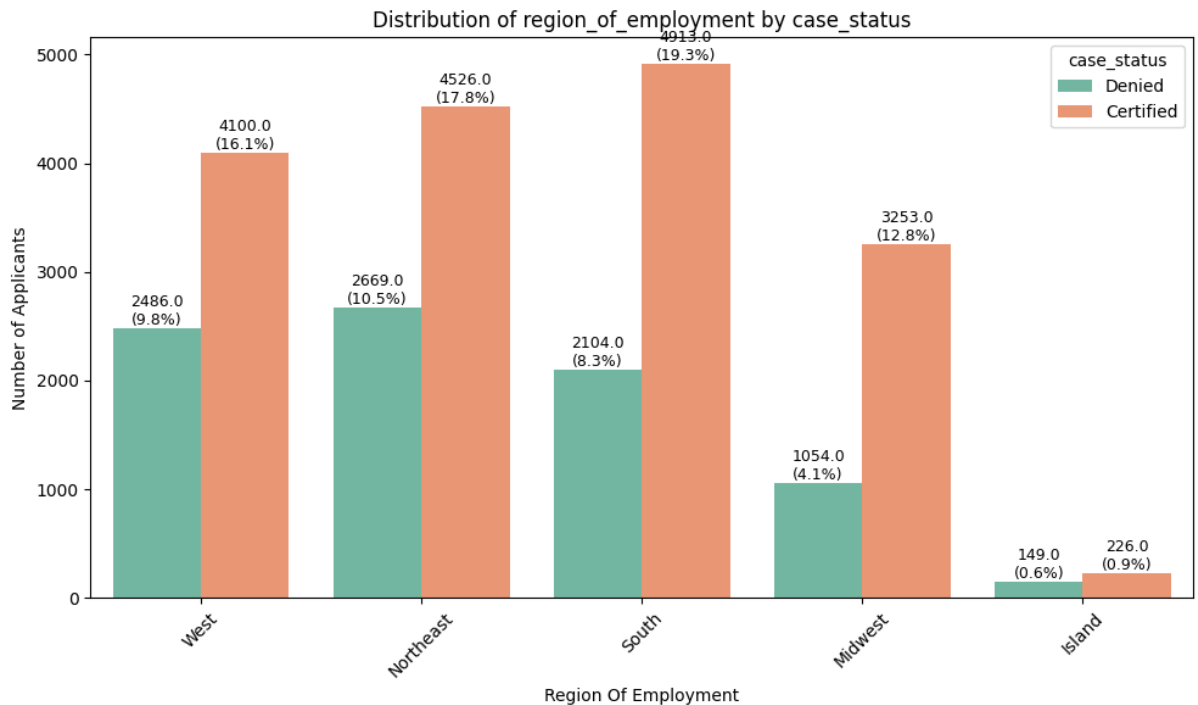


4. Observations on 'education_of_employee','case_status'



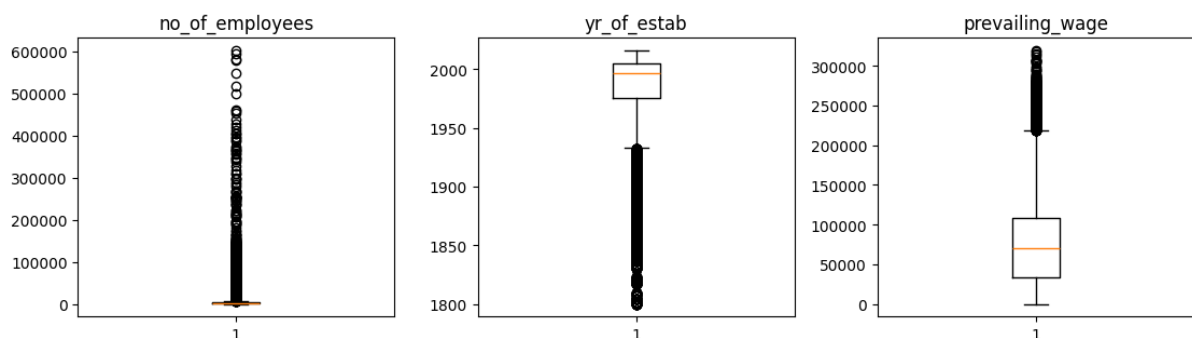
- Master's and bachelor's degree applicants are mostly certified. And amongst denied applicants, most of them are with bachelor's degree.

5. Observations on region_of_employment', 'case_status'



- Most of the certified applicants work in South and Northeast while Island Northeast region see the most number of visa denials.
- Least denials are seen in Island region

Outliers detection and treatment:



Observations

- There are quite a few outliers in the data.
- However, we will not treat them as they are proper values.

2. Defined features and Onehot-Encoding:

- we did onehot-encoding for all the categorical variables.
- And split the data to training, validating and testing data to check and perform the model building.

Results:

Shape of Training set : (15288, 10)

Shape of validating set:(5096, 10)

Shape of test set : (5096, 10)

- We then created dummies of all the categorical variables to analyse the data.

Model evaluation criterion:

We will be using Recall as a metric for our model performance because here there can be two types of problems

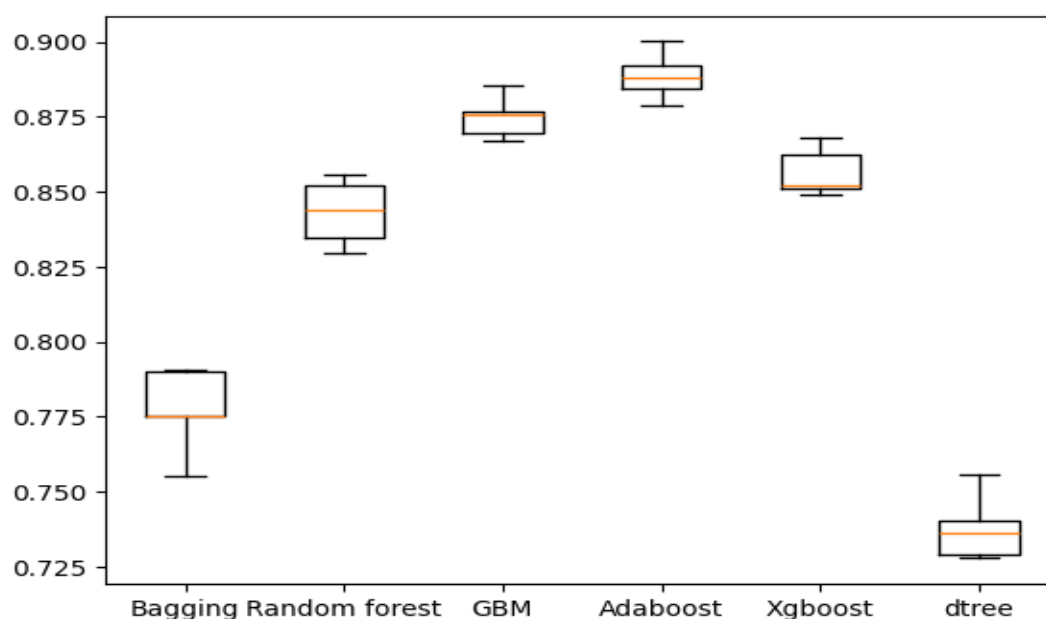
1. Applicants case_status can be denied
 2. Applicants case_status can be approved - this is needed to be studied. As our model need to identify applicants who are worthy of approval based on the other dependent variables.
- Metrics like recall and precision will then help assess how well we are finding truly qualifying applicants.
 - A high recall for 'approved' means we are catching most valid visa candidates — essential for reducing false denials.
 - A high precision for 'approved' ensures we are not approving those who actually shouldn't be.

Algorithm comparison:

Various models are compared and results are identified as below:

Cross-Validation Performance:	Validation Performance:
Bagging: 77.7179236043095	Bagging: 0.7705640423031728
Random forest: 84.30950048971596	Random forest: 0.8301997649823737
GBM: 87.48285994123408	GBM: 0.8742655699177438
Adaboost: 88.87365328109696	Adaboost: 0.8807285546415982
Xgboost: 85.6611165523996	Xgboost: 0.8522326674500588
dtree: 73.7806072477963	dtree: 0.745005875440658

Algorithm Comparison



- We can see that the Adaboost is giving the highest cross-validated recall followed by GBM & XG Boost.
- We will tune the best two models i.e. Adaboost & GBM and see if the performance improves.

Hyperparameter Tuning

We will tune Adaboost and GBM models using GridSearchCV and RandomizedSearchCV. We will also compare the performance and time taken by these two methods - grid search and randomized search.

First let's create two functions to calculate different metrics and confusion matrix, so that we don't have to use the same code repeatedly for each model

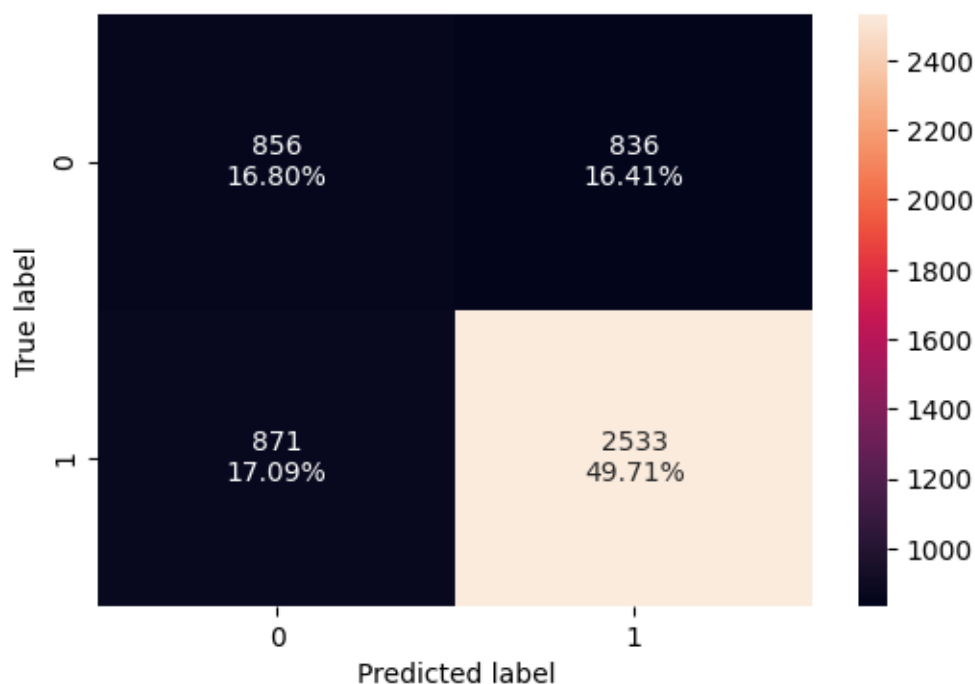
1. Adaboost – GridSearchCV

AdaBoost - Training performance:

	Accuracy	Recall	Precision	F1
0	1.000	1.000	1.000	1.000

AdaBoost - Validation performance:

	Accuracy	Recall	Precision	F1
0	0.665	0.744	0.752	0.748



- Accuracy is average in test data when compared with train data.
- Recall is good by 74% and precision is also good by 75% when compared to training data of 100%. This suggests Adaboost is learning well.

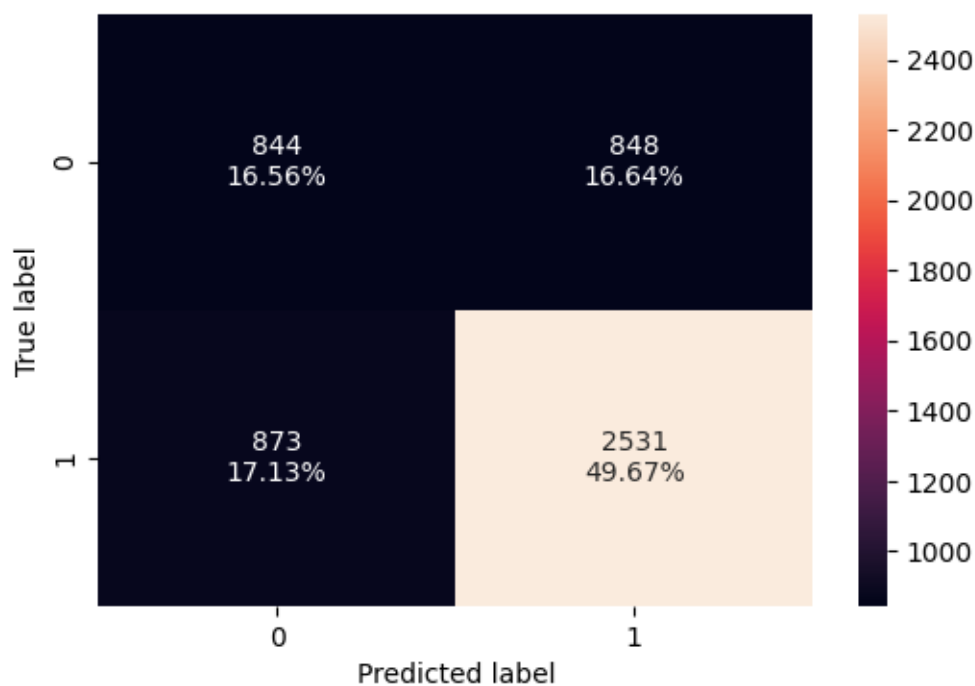
2. Adaboost - RandomizedSearchCV

Training performance:

	Accuracy	Recall	Precision	F1
0	1.000	1.000	1.000	1.000

Validation performance:

	Accuracy	Recall	Precision	F1
0	0.662	0.744	0.749	0.746



- Data is overfitting in training data.
- Confusion matrix still suggests numbers are very close — so about half of the truly approved applicants are being correctly classified, and the other half are slipping.

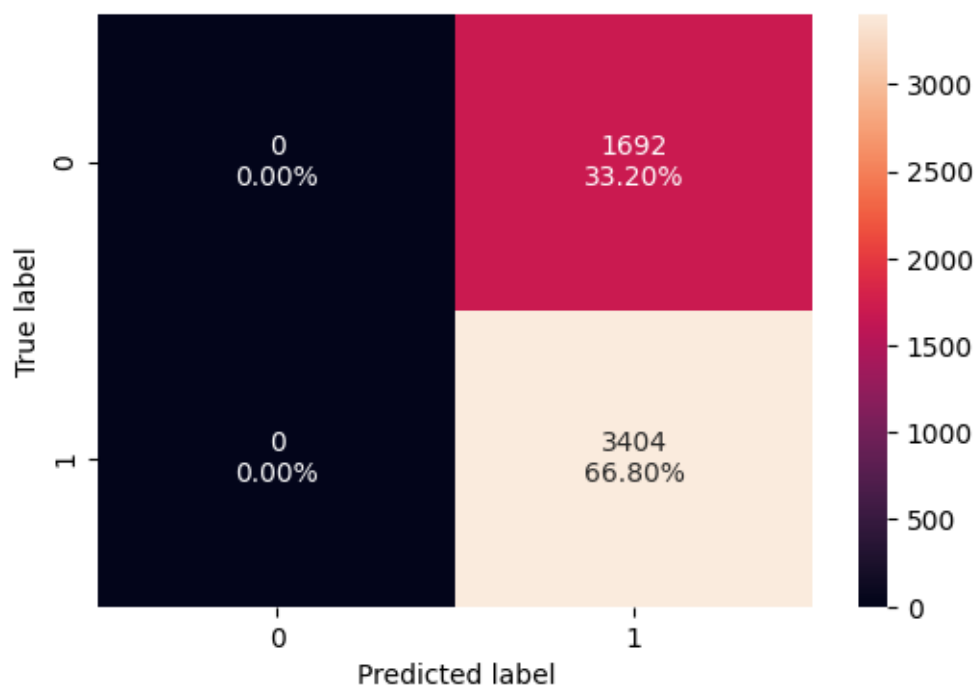
3.GBM – GridSearchCV

Training performance:

	Accuracy	Recall	Precision	F1
0	0.668	1.000	0.668	0.801

Validation performance:

	Accuracy	Recall	Precision	F1
0	0.668	1.000	0.668	0.801



- Recall is 1.000: It's correctly capturing all true positives.
- But Precision drops: Because it's predicting positives even when many are false.

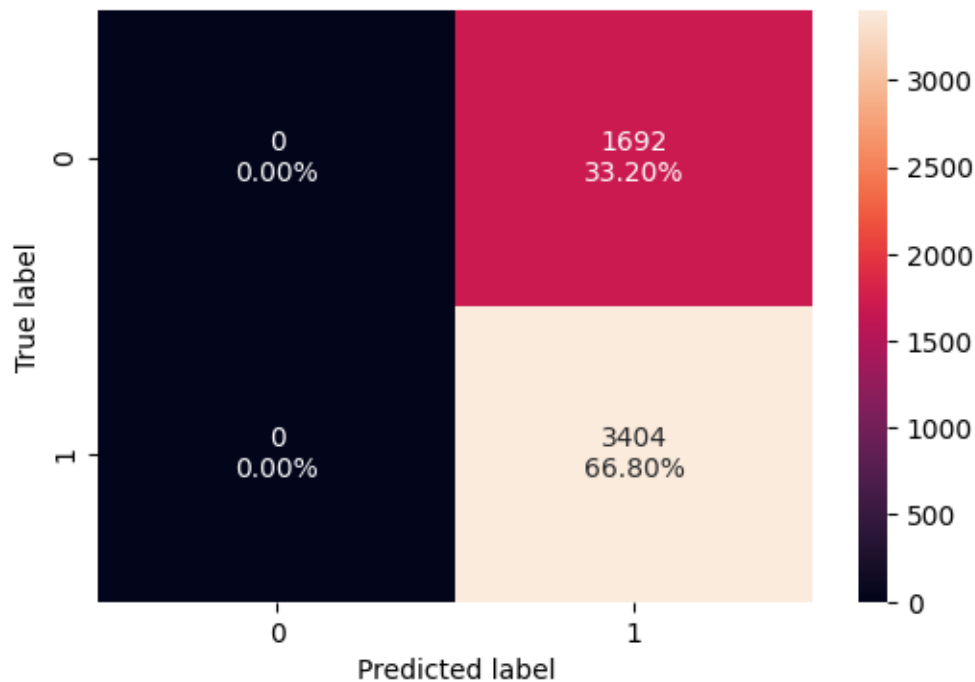
4.GBM – RandomizedSearchCV

Training performance:

	Accuracy	Recall	Precision	F1
0	0.668	1.000	0.668	0.801

Validation performance:

	Accuracy	Recall	Precision	F1
0	0.668	1.000	0.668	0.801



- model here shows perfect recall (1.000) but at the cost of a big false positive rate, which explains the modest precision (0.668) and the zero true negatives shown in the matrix.

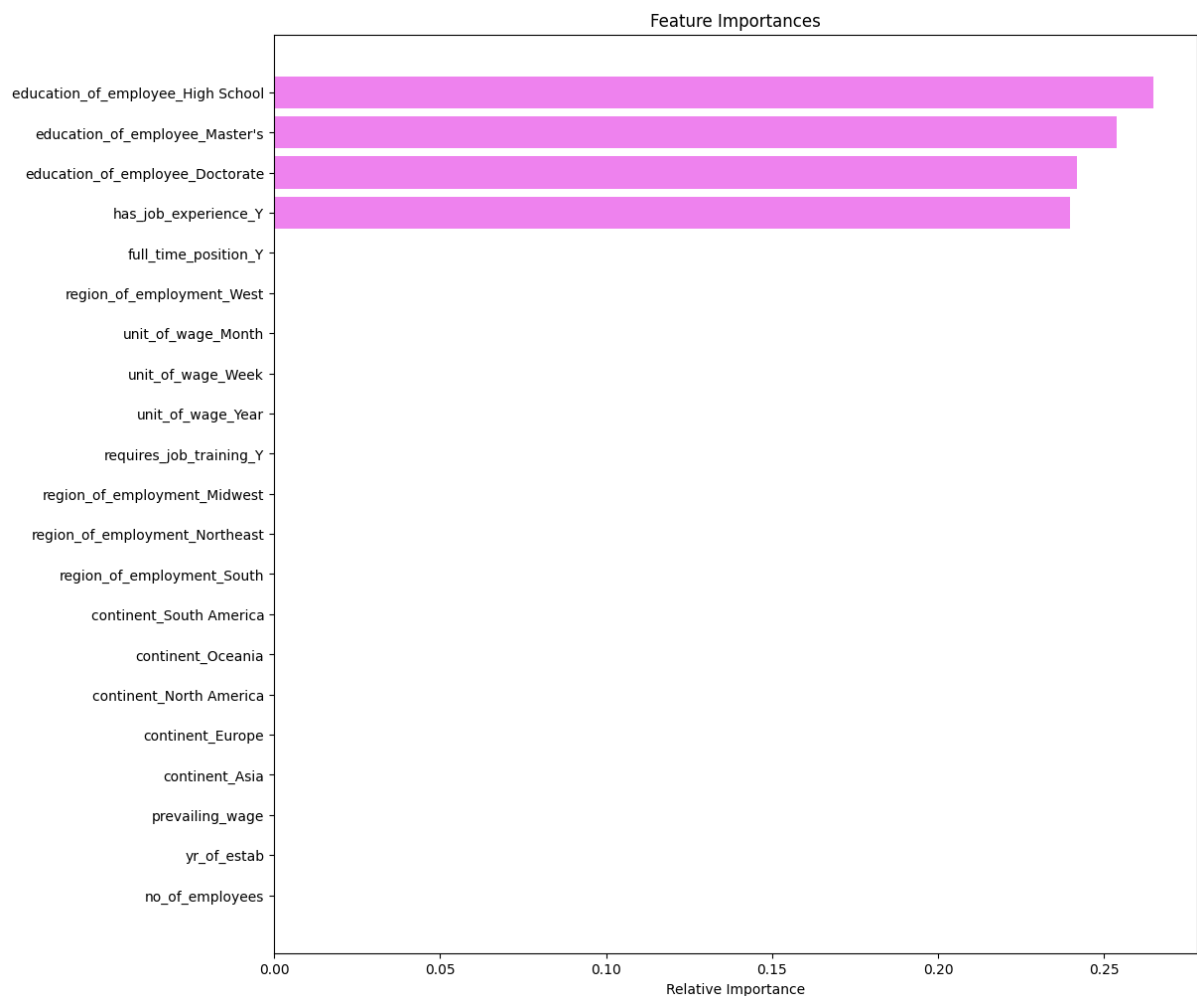
Comparing models from GridsearchCV and RandomisedsearchCV

- Adaboost models (both tuned by Grid and Random search) show perfect performance on training—this hints at possible overfitting.
- GBM models had lower accuracy and precision but still managed perfect recall, meaning they captured all true positives.
- Identical scores across search strategies (Grid vs Random) for each model type show consistent optimization effectiveness.

Key Insights

- Grid Search vs Random Search yielded similar outcomes across models, so Random Search might be preferable when time/resources are constrained.
- Adaboost might be overfitting—perfect training metrics but lower generalization.
- GBM, despite lower precision, maintains excellent recall and stable validation performance.

Feature Importance:



- Education and has job experience are the two most important variables as these variable play an important role in taking/returning credit.

Conclusion and Insights:

- GBM tuned via either method (Grid or Random Search) is your most balanced performer—excellent recall, strong F1, and stable across train/validation.
- Education and has_job_experience are the two most important variables as these variable play an important role in taking/returning credit.
- hence these points should be considered while selecting applicants.

-----THE END -----